S&P 500 Concentration Report: Returns, Top 10, and RSP/SPY

Oct 5, 2026 · @David Armendariz

Summary

Since January 2010 the S&P 500 has returned 14.27% a year with dividends, and its 10 largest companies added roughly 0.9 to 2.5 points of that each year. The market is now more concentrated than at any point since at least 2003.

Question	Answer (Jan 4, 2010 → Oct 5, 2026)
S&P 500 annualized return	12.18% price only, 14.27% with dividends
Without the top 10 companies	About 13.4% a year re-picking the top 10 each January; about 11.8% excluding today's top 10 throughout
Only the top 10, re-picked each January	17.27% a year: $10,000 became $144,138 vs $93,394 in the index, with a deeper worst drop (−39% vs −34%)
Equal-weight S&P 500 (RSP) vs cap-weight (SPY)	12.23% vs 14.15% a year
Is RSP/SPY a reliable warning signal?	No. It measures concentration well, but it has not timed corrections or rotations outside 2007–09

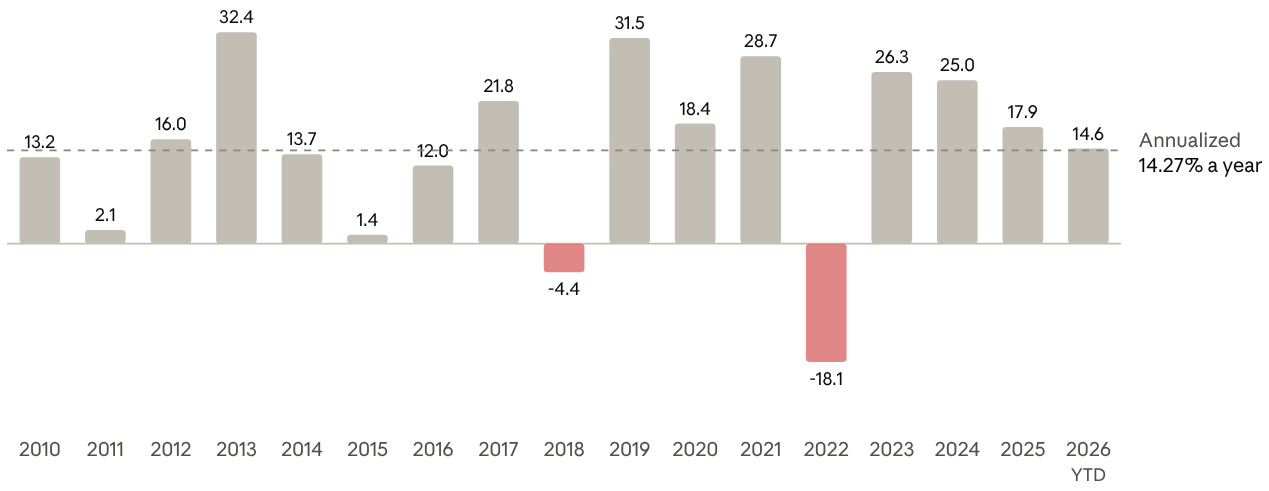
The top 10 now make up about 40% of the index, double their 2015 share. The RSP/SPY ratio is at its lowest level since RSP launched in 2003.

S&P 500 returns since 2010

The S&P 500 returned 14.27% a year with dividends reinvested, turning $10,000 into about $93,400 over 16.75 years. Price alone returned 12.18% a year, so dividends added about 2.1 points a year.

The S&P 500 rose in all but two years since 2010

Calendar-year return with dividends reinvested, %. 15 up years, 2 down years.



Yahoo Finance ^SP500TR · Jan 4, 2010 – Oct 5, 2026 · sp500_annualized_returns.ipynb, Part 1

Only 2018 and 2022 were down years. The worst peak-to-trough drop was −33.8%, in the March 2020 crash.

The S&P 500 without its top 10

Without its 10 largest companies, the S&P 500 would have returned about 0.9 to 2.5 points less a year since 2010, depending on how the top 10 are picked. No such index is published, so it was rebuilt from constituent data (see Method).

Version (dividends included)	Annualized	vs rebuilt full index
Rebuilt full index	15.46%	—
Without each year's top 10, re-picked every January	14.61%	−0.85 pts
Without today's top 10, for the whole period	13.00%	−2.46 pts
Today's top 10 only	23.82%	+8.35 pts

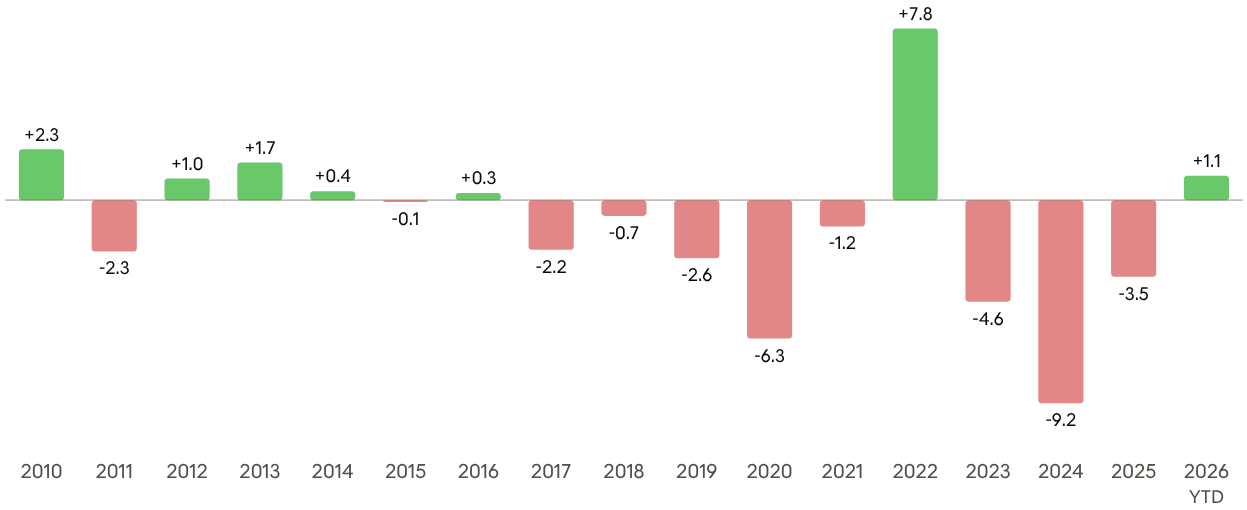
The rebuilt full index runs 1.19 points a year above the official 14.27%, mostly because companies that left the index since 2010 are missing. Read the gaps, not the levels. Applied to the official index, they put the S&P 500 without its top 10 at about 13.4% (re-picked yearly) or 11.8% (today's top 10).

Re-picking each January is the fairer test. Removing today's top 10 builds in hindsight, since those companies are the largest *because* they rose the most.

Without its top 10, the index lagged most in 2020 and 2023–2025

Yearly return of the index without that year's 10 largest companies, minus the full index, in points. Behind in 10 of 17 years.

Green: the index did better without the top 10. Red: the top 10 lifted the index.



Rebuilt from today's constituents, Yahoo Finance prices and share counts · Jan 4, 2010 – Oct 5, 2026 · notebook Part 2

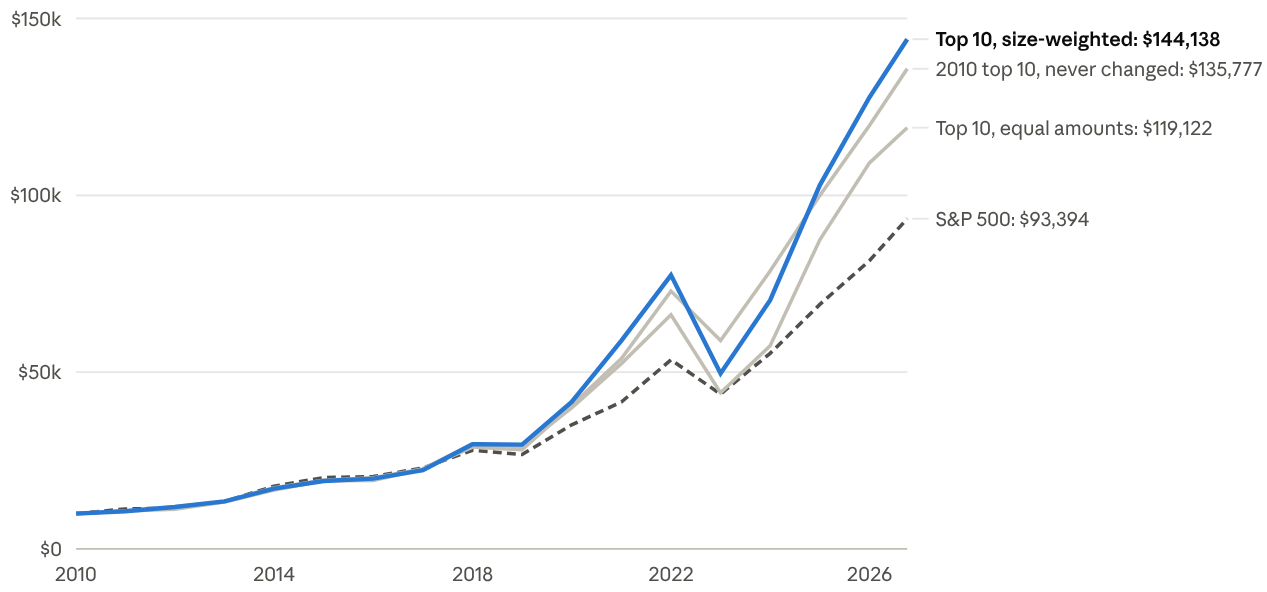
The top 10's contribution is recent. Before 2017, the index with and without them moved roughly together. In 2022 the giants fell harder, and the rest of the index beat them by 7.8 points. Today the top 10 make up about 40% of the index: Nvidia, Apple, Alphabet, Microsoft, Amazon, Meta, Broadcom, Tesla, Micron and Berkshire Hathaway.

Investing only in the top 10

Buying the 10 largest S&P 500 companies each January and holding them for the year returned 17.27% a year, 3 points more than the index. It did so with more risk, so the return per unit of volatility was about the same (0.85 vs 0.83).

$10,000 in each January's top 10 became $144,138, vs $93,394 in the S&P 500

Value of $10,000 invested Jan 4, 2010, dividends reinvested, year-end values



Rebuilt from Yahoo Finance prices and share counts; S&P 500 = ^SP500TR · Jan 4, 2010 – Oct 5, 2026 · notebook Part 3

Strategy (Jan 4, 2010 → Oct 5, 2026)	Annualized	Volatility	Worst drop	Return / volatility	Years beating S&P
S&P 500 (dividends included)	14.27%	17.2%	−33.8%	0.83	—
Top 10, size-weighted, re-picked each January	17.27%	20.2%	−39.2%	0.85	11 of 17
Top 10, equal amounts, re-picked each January	15.94%	19.1%	−36.5%	0.83	10 of 17
2010 top 10, bought once and never changed	16.85%	17.5%	−28.6%	0.96	12 of 17

- **The edge only appeared after 2015.** From 2010 to 2014 the size-weighted top 10 returned 14.0% a year vs 15.1% for the index. The gap opened in 2015–2019 (16.7% vs 11.7%) and widened from 2020 (20.2% vs 15.6%).
- **2022 was the cost.** The size-weighted top 10 fell 35.9% vs 18.1% for the index, a 17.8-point shortfall.
- **Size-weighting beat equal amounts** because it let the biggest winners (Apple, Microsoft, Nvidia) grow into a larger share.

- **The 2010 list held up well** with the smallest worst drop. It mixed Apple, Microsoft and Google with steadier names like Exxon, AT&T, Walmart and J&J. It is a single list, so treat it as one example, not a rule.

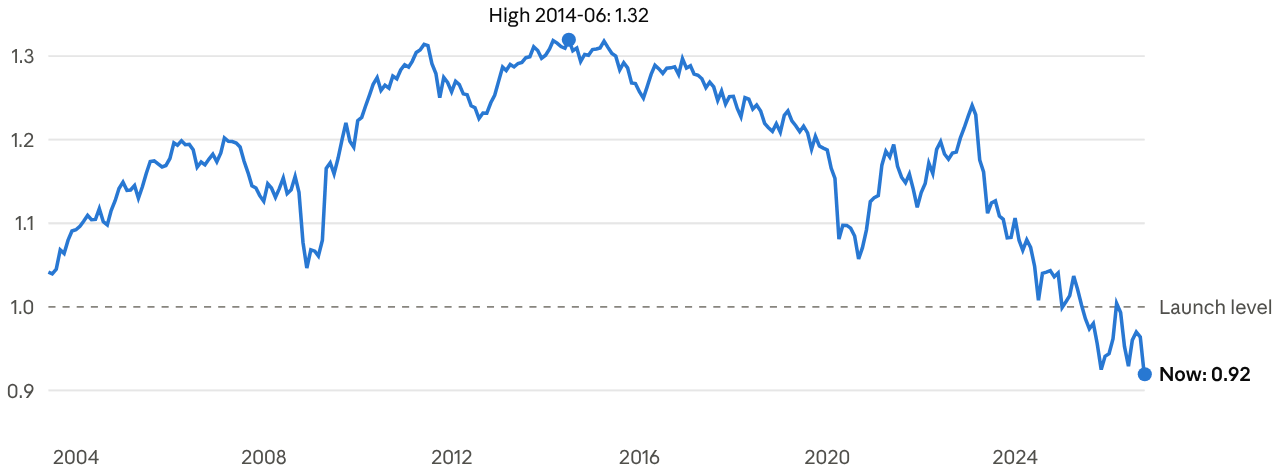
On average 1.4 of the 10 names changed each January. Taxes and trading costs are ignored.

RSP vs SPY: is the ratio a reliable signal?

The RSP/SPY ratio is a good gauge of how concentrated the market is, but it has not been a reliable timing signal. RSP holds the same stocks as SPY at about 0.2% each, so the ratio tracks the average stock against the cap-weighted index. Since 2010 RSP returned 12.23% a year vs 14.15% for SPY.

Equal weight vs cap weight is at its lowest level since 2003

RSP/SPY total-return ratio, month-end, May 1, 2003 = 1. Now 30% below its month-end high.



Rising line: the average stock beat the index. Falling line: the largest companies led.

Yahoo Finance RSP and SPY, dividend-adjusted · May 1, 2003 – Oct 5, 2026 · notebook Part 4

The statement under test makes four claims. Each was checked against RSP and SPY prices since 2003 and the constituent data from 2010.

Claim	Verdict	Evidence
Rising ratio = broad, healthy market; falling = mega-caps lead	Half true	It measures the average stock against the giants. But the ratio also rose in 2009 and 2022, a crash rebound and a bear market. It is only loosely tied to real breadth: +0.33 correlation with changes in the share of stocks above their 200-day average.
It exposes concentration in the top 7–10 stocks	True	It moves almost exactly opposite to the top-10 weight: −0.86 correlation in 3-month changes. The top-10 weight doubled from about 20% in 2015 to about 40% today.
A low or falling ratio warns of a correction	Not supported	The warning comes almost entirely from 2007–09. Outside that period, the chance of a >10% drop was the same after a weak ratio as on any day. The S&P 500's next-12-month return was higher.
A breakout signals smaller companies taking the lead	Weak	Ratio moves barely persist: autocorrelation +0.11 monthly, +0.06 quarterly. After a 52-week high the ratio was higher 6 months later 62% of the time, but slipped on average. Crossing its 200-day average predicted nothing.

The correction test, split by period. A weak ratio means its 6-month change was in the bottom 20% of all days since 2003:

Period	SPY next 12 months, all days	SPY next 12 months, after a weak ratio	Chance of a >10% drop in 6 months, all days	Same, after a weak ratio
2004–2009 (includes 2008)	4.3%	3.0%	31.0%	86.9%

Period	SPY next 12 months, all days	SPY next 12 months, after a weak ratio	Chance of a >10% drop in 6 months, all days	Same, after a weak ratio
2010–2019	13.5%	17.4%	30.4%	36.4%
2020–now	17.7%	23.6%	29.2%	21.5%
Whole history minus 2007–2009	14.6%	21.7%	25.7%	25.6%

The ratio has fallen since 2014–15 while the S&P 500 kept rising. Use it to see how concentrated your S&P 500 exposure is, not to time pullbacks. Holding RSP as a diversifier is a bet that the trend reverses, and since 2010 that bet has cost about 1.9 points a year.

Method and caveats

All figures come from the notebook `sp500_annualized_returns.ipynb`, run on Oct 5, 2026 with Python 3.14. Returns include reinvested dividends unless marked as price only. Taxes and trading costs are ignored.

- **Index data**: Yahoo Finance `^GSPC` (price) and `^SP500TR` (total return), and ETFs `RSP` and `SPY` (dividend-adjusted).
- **Rebuilt index**: today's 503 tickers (Wikipedia constituent list), one ticker per company, weighted by estimated market cap. Market cap = price × shares outstanding, using Yahoo's share-count history adjusted for stock splits.
- **Survivorship bias**: companies that left the index since 2010 are missing, so the rebuilt index runs 1.19 points a year above the official one. Comparisons use gaps against the rebuilt index, which cancels most of this. The top-10 strategy is barely affected: every company that made a yearly top 10 is still in the index.
- **Share counts before 2015**: Yahoo's history starts around 2015, so earlier weights reuse each company's earliest count. Top-10 picks for 2010–2014 may be slightly off.
- **Weighting**: the official index uses float-adjusted market caps, which the rebuild does not.
- **No look-ahead**: each year's top 10 are picked at the close of the first trading day and held from the next day.

- **Overlapping windows** (RSP/SPY tests): forward returns from nearby days overlap, so there are far fewer independent episodes than days. Splitting by period guards against one episode dominating.